

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 04
Affine Functions.

Review

1. (**Notation**) What is $\mathbb{R}$?
2. (**Notation**) What is $\mathbb{R}^n$?
3. What is an *additive* function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
4. What is an *homogeneous* function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
5. What is an *linear* function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
6. Show that the function $f(x, y) = 2x + 3y$ is linear.
7. Any linear function can be written as an inner product.

New (from Chp.2)

1. How can we write the linear function $f(x, y) = 2x + 3y$ as an inner product?
2. Why do all linear functions go through the origin? (If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear function, show $f(\vec{0}) = 0$).
3. Why is the function $f(x, y) = 2x + 3y + 5$ *not* linear?

4. Why is the function $f(x, y) = 2x + 3y + 5$ an *affine* function?

Norm and Distance (Chp. 3)

5. What is the *norm* of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$?

6. What does this “norm” correspond to our regular notion of distance in $\mathbb{R}^2$?

7. What is the RMS of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$?